



QUANTUM PNT FOR DRONES

UC-085 • DEFENCE & STRATEGIC APPLICATIONS

VOID-NAV

Navigating without the sky: a quantum-assisted positioning payload that reads the Earth's magnetic fingerprint when GNSS is jammed, spoofed or unavailable.

Quantum sensing

Grover search

Quantum kernel ML

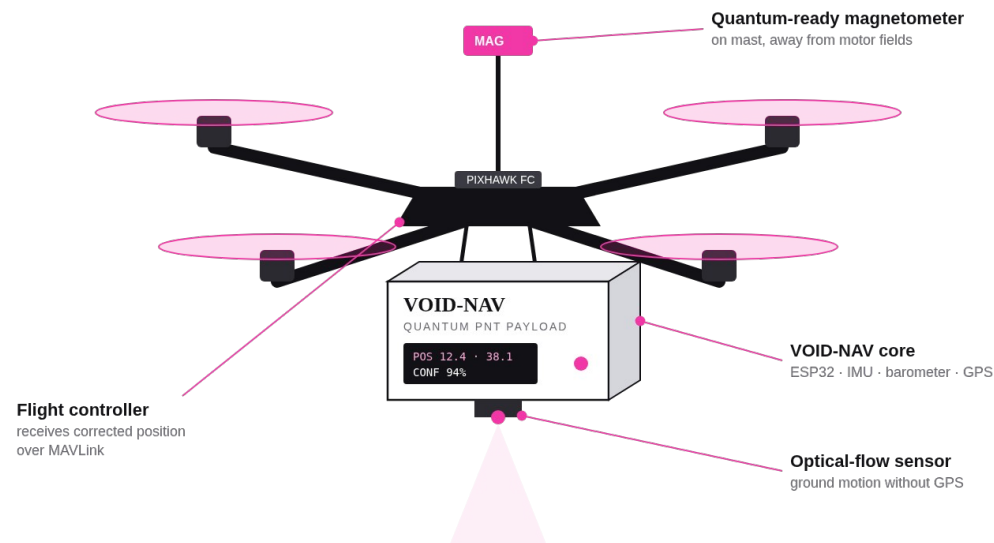
QAOA planning

TEAM

Team VOID

DELIVERABLE

Working payload prototype + Qiskit pipeline + live dashboard



VIDEO PITCH

<https://youtu.be/BV2KYV-yhL0>



PROBLEM STATEMENT

Without GNSS, a drone's inertial position drifts without bound.

10^{-16} W

GNSS power at the receiver: weaker than background noise, easily denied by jamming, terrain, tunnels or foliage.

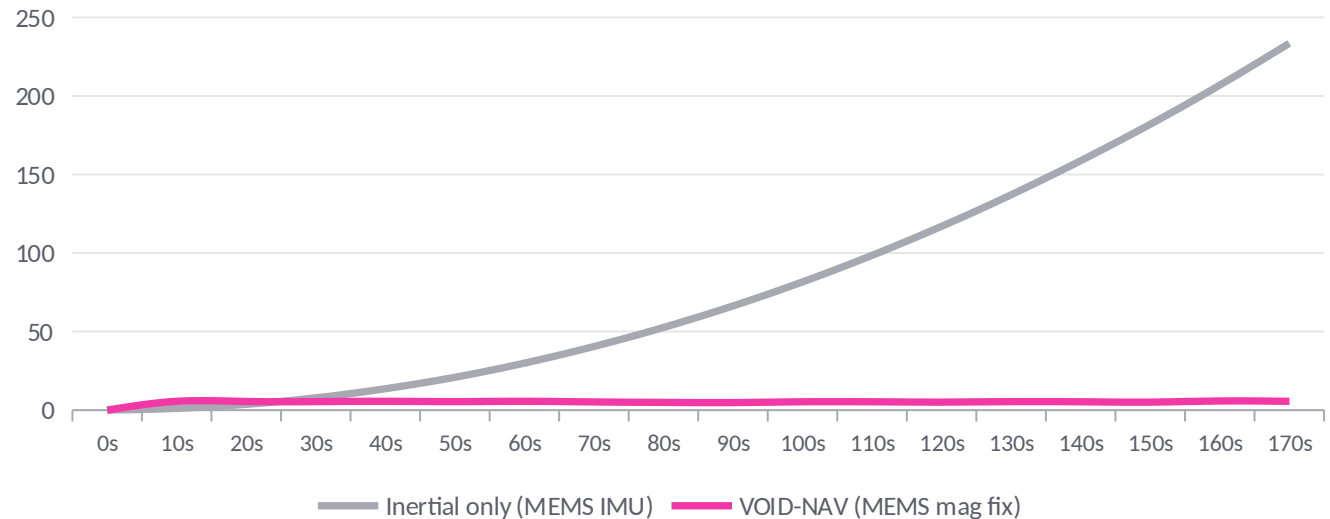
~260 m

Mean error after 3 min of MEMS-only dead reckoning in our drift model: error grows with t^2 , not t .

₹10L+

Cost of navigation-grade INS or cold-atom units: too expensive to fit on every tactical drone.

Position error after GPS loss (m), simulated

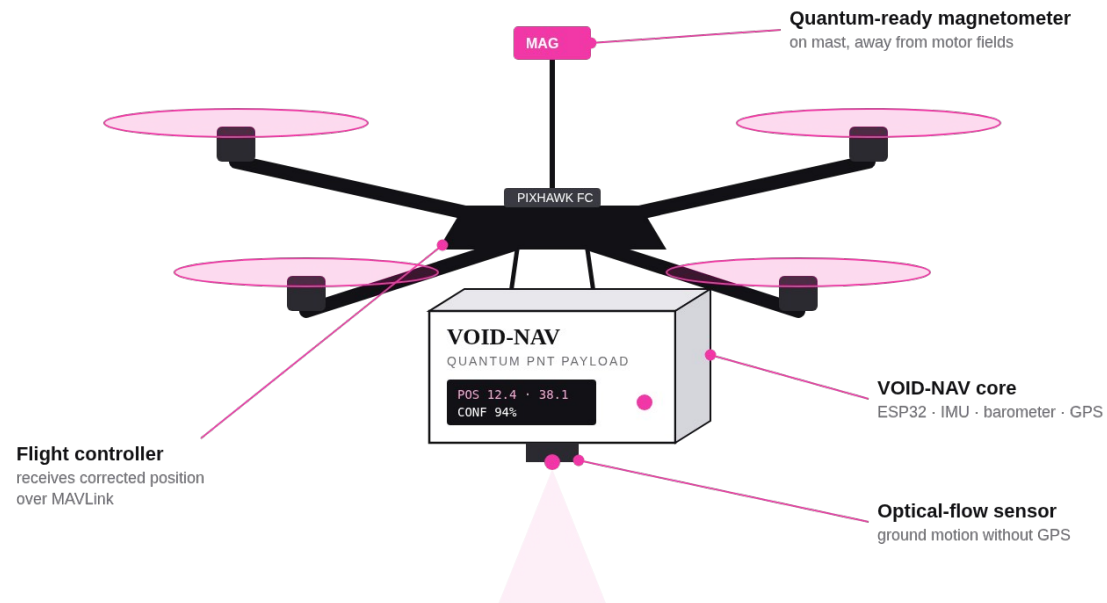


Problem: give a defence UAV a position it can trust within metres when GNSS disappears, with a passive (non-emitting), low-cost payload that is ready to take a quantum magnetometer, as UC-085 and the iDEX passive quantum positioning challenge call for.



PROPOSED SOLUTION

VOID-NAV: a passive quantum PNT payload. The drone carries it; the payload does the navigation.



1

Map

GPS on: fly or walk the area once and log a 5-D magnetic fingerprint [Bx, By, Bz, |B|, ∇B] per 2 m cell.

2

Sense

GPS off: the magnetometer, IMU, optical flow and barometer track motion; the magnetic field is read as a short window.

3

Match

Quantum engine: Grover's search finds candidate cells, the quantum kernel SVM confirms the zone.

4

Correct

A particle filter fuses the match with dead reckoning, resets IMU drift and sends position + confidence to the autopilot.



INNOVATION & UNIQUENESS

Most entries show one quantum algorithm. VOID-NAV runs a complete quantum PNT pipeline.

01 · FULL STACK

Sense, search, learn, plan

Quantum sensing, Grover's search, quantum-kernel ML and QAOA each do a distinct job in one navigation loop.

02 · RICHER FINGERPRINT

5-D magnetic gradient map

We match the spatial pattern $[B_x, B_y, B_z, |B|, \nabla B]$, not a single field value, which cuts false matches sharply.

03 · QUANTUM CROSS-CHECK

Grover proposes, QSVM decides

In our run Grover returned two look-alike cells (0110, 1011); the QSVM zone label resolved the tie.

04 · QUANTUM-SENSOR READY

Drop-in NV-diamond slot

Same data interface as an NV magnetometer; we simulate its Ramsey readout today to quantify the upgrade.

05 · MAGNETICALLY AWARE ROUTES

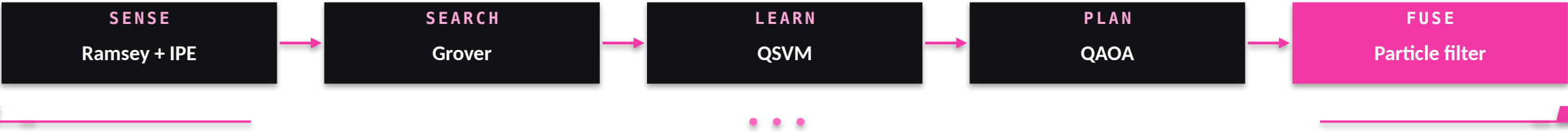
QAOA path planning

Chooses waypoints through magnetically rich cells so the drone gets more position fixes per minute.

06 · DEPLOYABLE

Passive, retrofit, low cost

Emits nothing, speaks MAVLink to any ArduPilot/PX4 drone, prototype under ₹8,000 in parts.



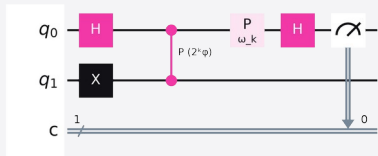


QUANTUM COMPUTING APPROACH

Four quantum building blocks, each mapped to one navigation job.

Q1 • QUANTUM SENSING

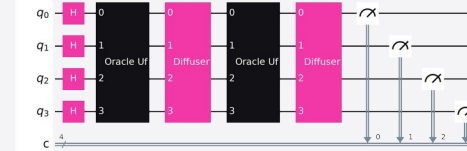
Ramsey interferometry + iterative phase estimation



Qubit in superposition via H; field B imprints phase $\phi = \gamma B \tau$ ($\gamma \approx 28$ GHz/T); phase kickback + IPE reads ϕ bit by bit. Models an NV-diamond readout.

Q2 • SEARCH

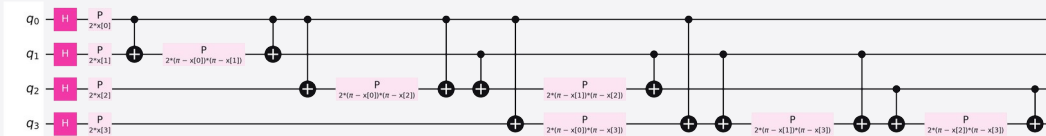
Grover's algorithm • 4 qubits, 16 map cells



Oracle U_f phase-flips cells whose fingerprint matches; diffuser $2|s\rangle\langle s| - I$ amplifies them. $\sim\sqrt{N}$ queries; 2 iterations here.

Q3 • LEARNING

Quantum-kernel SVM • ZZ feature map



$x \mapsto |\phi(x)\rangle$ in a 16-dim Hilbert space; kernel $K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$ feeds a classical SVM (QSVC) to label the zone.

Q4 • PLANNING

QAOA • $p = 1$, route graph



Cost Hamiltonian $H_C = \sum w_{ij} Z_i Z_j$ rewards magnetically rich waypoints; mixer $H_M = \sum X_i$; γ, β tuned by COBYLA.



QISKIT & TECHNOLOGY STACK

Open-source software on borrowed hardware: built for a student budget.

QUANTUM	Qiskit 2.x SDK · Qiskit Aer · IBM Runtime SamplerV2 · qiskit-machine-learning (FidelityQuantumKernel, QSVC) · QAOA + COBYLA
NAVIGATION	Python · NumPy / SciPy · FilterPy particle filter + EKF · hard/soft-iron calibration
FLIGHT	ArduPilot SITL · pymavlink · MAVLink VISION_POSITION_ESTIMATE / GPS_INPUT
FIRMWARE	ESP32 · Arduino C++ · I ² C sensor drivers · 50 Hz UDP stream
INTERFACE	Streamlit + Plotly live map · circuit + histogram panels · CSV / Parquet logs

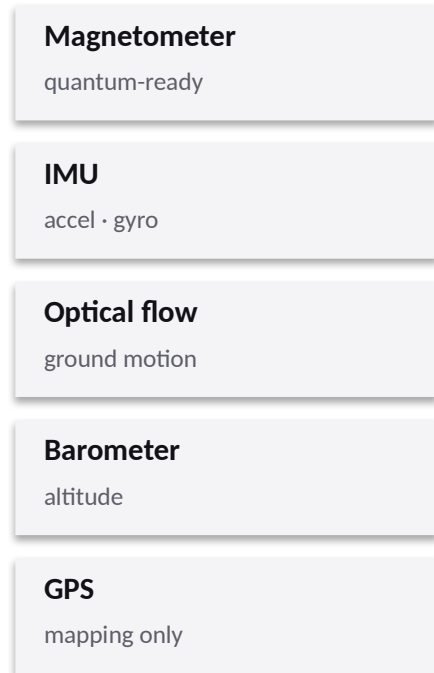
Part	Role	Source	₹
QMC5883L magnetometer	Main field sensor (mast)	Buy	300
MPU-9250 IMU	Accel + gyro	Borrow	600
BMP280 barometer	Altitude	Borrow	150
NEO-6M GPS	Mapping + ground truth	Borrow	450
PMW3901 optical flow	Ground motion	Buy (opt.)	2,000
ESP32 DevKit	Sensor hub	Borrow	450
Quadcopter frame / drone	Platform	Borrow	0
Total if bought			≈ 3,950



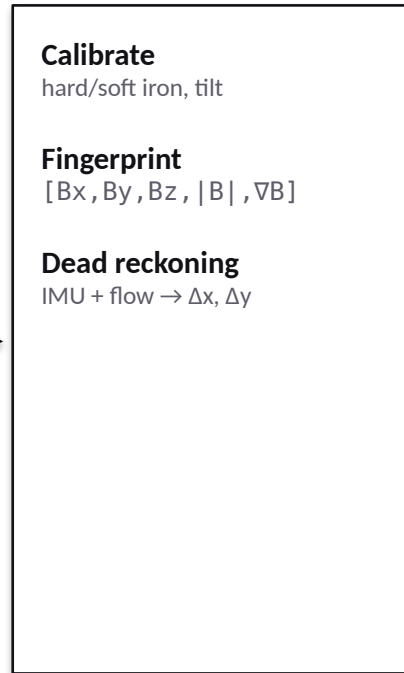
SYSTEM ARCHITECTURE / WORKFLOW

Sensors in, trusted position out: quantum where it adds value, classical where speed matters.

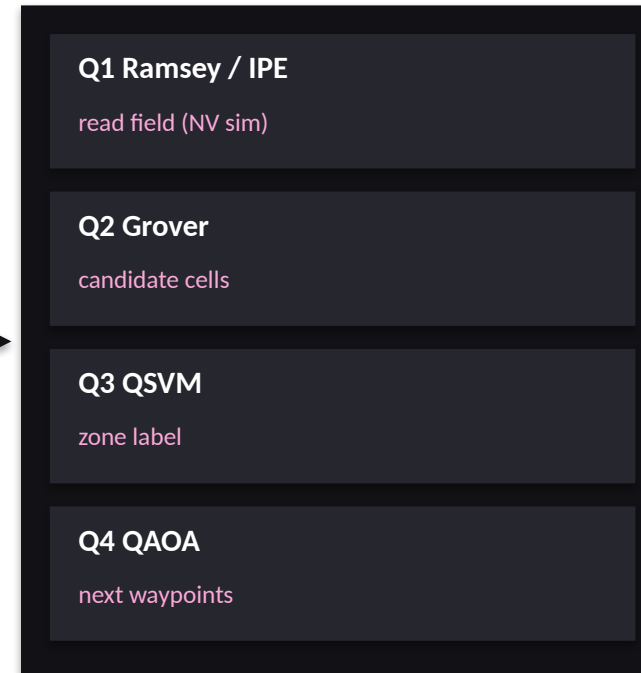
SENSE · 50 HZ



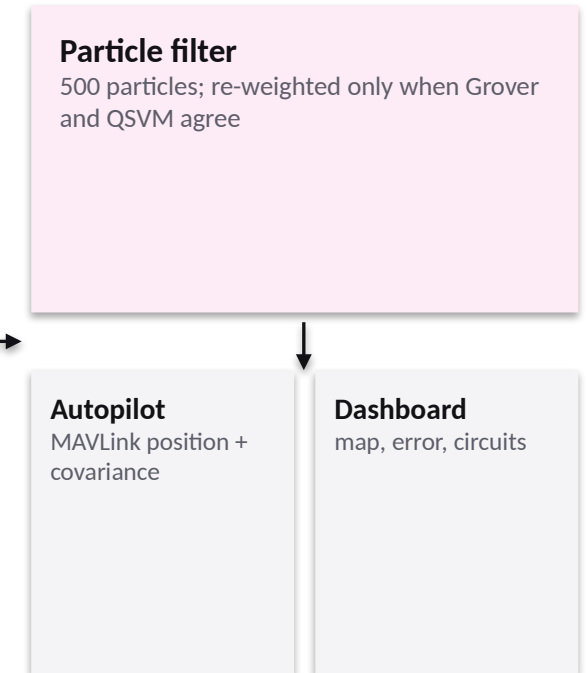
PRE-PROCESS



QUANTUM ENGINE · QISKIT



FUSE & ACT



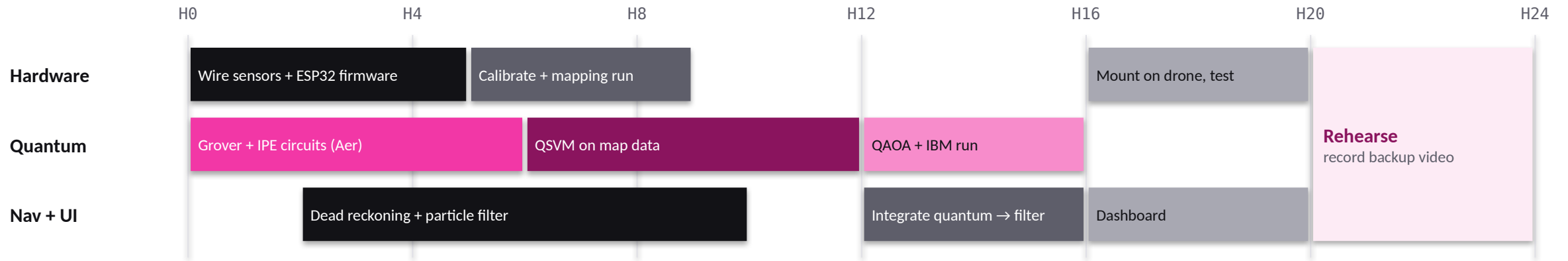
Loop: classical fusion runs every 20 ms; quantum matching runs every 1 s on a window of readings, so quantum latency never blocks flight control.



IMPLEMENTATION & FEASIBILITY

Built in 24 hours, with a working fallback at every level.

24-HOUR BUILD PLAN



Level 1 · guaranteed

Payload carried along the mapped corridor; live dashboard shows drift vs correction.

Level 2 · main demo

Payload mounted on a borrowed drone, carried or tethered; full flight replayed in ArduPilot SITL on our real map.

Level 3 · bonus

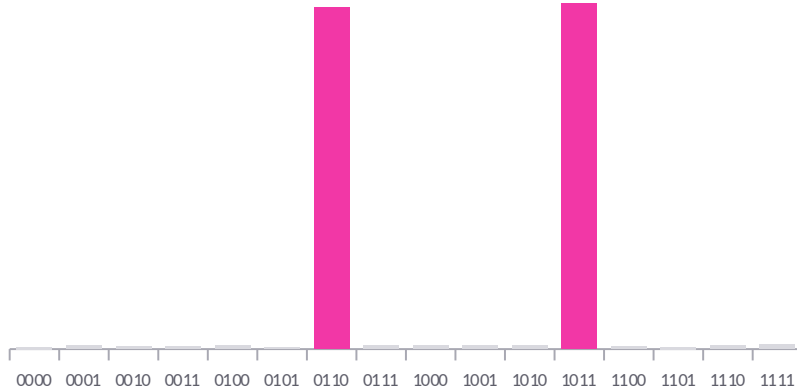
Short real flight with GPS disabled in software, only if a flight-ready drone and venue permission exist.



EXPECTED RESULTS & IMPACT

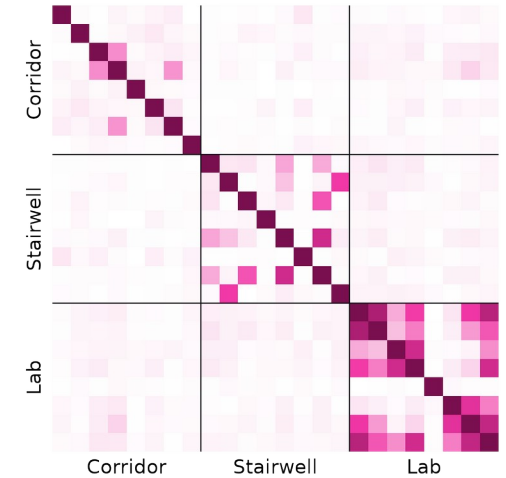
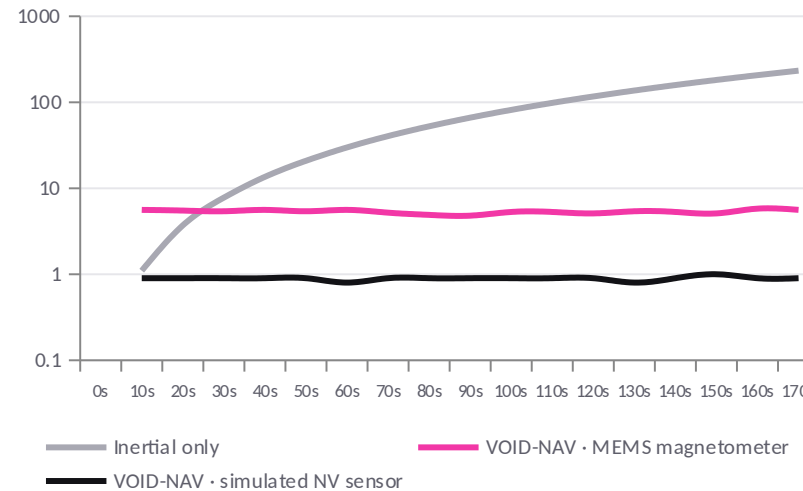
Early results from our own Qiskit runs and navigation simulation.

Grover on Aer - 4,096 shots



0110 + 1011 = 94% of shots. Grover returns 2 look-alike cells; QSVM breaks the tie.

Error after GPS loss (m, log scale)



Quantum kernel matrix. Same-zone fingerprints overlap $\sim 2.8\times$ more than cross-zone ones (0.24 vs 0.09).

$\sim 50\times$

lower drift vs inertial-only (MEMS model)

$\sim 6\times$

further gain with a quantum (NV) magnetometer

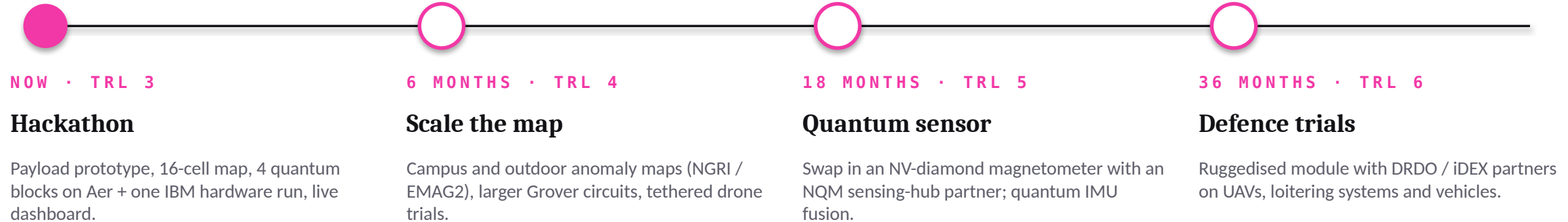
0 W

radio emission: fully passive



FUTURE SCOPE & CONCLUSION

From a hackathon payload to a field-ready quantum PNT module.



"When GPS goes dark, the ground still knows where you are."

VOID-NAV: quantum sensing, quantum search, quantum learning and quantum planning, flying on one passive payload.

Team VOID

THANK YOU · QUESTIONS WELCOME

Video: <https://youtu.be/BV2KYV-yhL0>